

WESTFIELD CONCEPT MANUFACTURING BRIEF · DRAFT

Insulated Concertina Cooler Manufacturing Strategy

A practical construction path for an expandable cooler with a faceted external structure: make the outside carry the visual fold language, then solve insulation, leak resistance, and load-bearing with independent layers.

Recommended Direction

Do not treat the entire wall as one continuous insulated accordion. Thick foam and waterproof liners do not like repeated sharp creasing. The stronger path is a semi-rigid segmented soft cooler: faceted outside shell, separate insulated panels, controlled hinge valleys, welded liner, and a load path that bypasses the folding wall.

BEST ARCHITECTURE

Segmented insulated wall

Put closed-cell foam panels on the broad faces and leave the valleys as thin flex zones. This keeps insulation where it matters without crushing foam every time the cooler collapses.

VISUAL STRATEGY

Faceted external exoshell

The orange concertina geometry should be the outer shape/form layer. It can be laminated fabric, TPU-coated textile, tarpaulin, or thin thermoformed EVA/TPU panels.

LEAK STRATEGY

Welded inner bucket

Use a separate TPU or PEVA liner with slack for expansion and collapse. Heat, RF, or induction welding is preferable to sewn liner seams for a leak-resistant cooler story.

CRITICAL RULE

Carry load outside the folds

Handles and shoulder straps should connect to reinforced rails, side columns, rim, or base. The folded wall should shape volume, not carry the full ice and beverage load.

Proposed Construction Stack

The concept works best as a three-layer bag plus independent carry structure. This separates visual form, insulation, waterproofing, and load-bearing so each material can do the job it is good at.

Faceted outside, hidden thermal system



Rigid/semi-rigid rim
Stabilizes opening and zipper or roll-top alignment.

Faceted outer shell
Creates the concertina look and crush-resistant form.

Load rails
Carry straps bypass the folding wall.

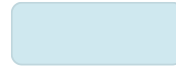
Stable base
Prevents loaded cooler sag and twist.



Outer shell: TPU-coated polyester/nylon, laminated tarpaulin, heavy 600D-1680D polyester, or thin thermoformed EVA/TPU panels heat-welded or sewn into zig-zag faces.



Insulation: EPE/XPE/closed-cell foam panels on broad wall faces. Use thinner flexible hinge strips at valleys. Aerogel fabric is a premium experiment for thin cold-bridge zones.



Inner liner: Welded TPU or PEVA bucket liner, slightly oversized so it can expand and collapse without pulling at seams or corners.



Load structure: Webbing rails, side columns, reinforced base, and rim carry the weight. Folding facets remain a shape/volume system.

Manufacturing Options

The right construction depends on target price and performance. These routes are ordered from most practical near-term path to higher-risk premium exploration.

Route	How it is made	Why it matters
Soft panel cooler	TPU-coated or polyester shell, EPE/XPE foam panels, welded liner, webbing reinforcement, sewn/welded perimeter.	Most manufacturable. Fits existing soft cooler suppliers and gives a realistic cost/performance starting point.
Faceted shell + foam insert	Outer zig-zag shell is formed separately; segmented foam/liner assembly nests behind it.	Best way to make the external structure expressive without forcing every layer to accordion together.
Thermoformed EVA/TPU facets	Thin molded panels or laminated panels are attached to flexible hinge zones.	Could feel more structural and premium, but tooling and crease durability need validation.
Thin-insulation hinge upgrade	Flexible aerogel blanket/fabric, felted insulation, or coconut fiber fills hinge zones where foam cannot fold.	Useful if thermal testing shows cold bridges at the valleys. Likely higher cost and sourcing complexity.

Risks To Solve Early

The design risk is not whether a collapsible cooler can be made. It is whether the folded exterior can stay clean, strong, insulated, leak-resistant, and easy to collapse after real use.

THERMAL

Cold bridges at hinge valleys

Thin fold zones will underperform compared with foam faces. Prototype with and without thin flexible insulation in the valleys and test ice retention.

WATER

Valleys trap condensation and dirt

External valleys need drain-friendly geometry and wipeable materials. Avoid deep seams that collect grime or stay wet after use.

DURABILITY

Foam fatigue and crease memory

Repeated sharp folding can crush foam and weaken laminations. Keep foam off the hinge line and cycle-test collapse/open motion.

USABILITY

Liner slack and zipper alignment

The inner bucket needs controlled slack so it does not pinch, pull, or block closure in collapsed and expanded states.

STRUCTURE

Loaded sidewall distortion

Ice and drinks can distort flexible folded walls. Handles should anchor into the base/rim/reinforced rails, not just the outer facets.

COST

Premium material creep

Aerogel and specialty laminates may solve thin-zone insulation, but they can quickly push the design away from a practical retail target.

Prototype Plan

Start with a low-risk construction mockup that proves the product behavior before investing in molded panels or expensive insulation.

Build first

- Make a 4-6 panel wall mockup with XPE/EPE foam on broad faces.
- Use TPU-coated or tarpaulin outer fabric with fabric-only hinge valleys.
- Add a removable or welded oversized liner bucket.
- Add compression straps, snaps, or hooks to hold collapsed, mid, and full states.
- Use a rigid/semi-rigid base and rim to control loaded shape.

Test immediately

- Ice retention against a normal soft cooler of similar volume.
- Leak and condensation behavior after meltwater exposure.
- Open/collapse cycle durability for hinge valleys and foam panels.
- Loaded carry test to confirm straps bypass the folding wall.
- Cleaning test for the external valley geometry.

VERDICT

Best manufacturable concept

A semi-rigid segmented soft cooler with an expressive faceted shell. Make it look like a concertina, but let panelized foam, a welded liner, and independent webbing do the real performance work.

PREMIUM EXPLORATION

Where to experiment later

Evaluate flexible aerogel fabric, coconut fiber insulation, or felted hinge insulation only after the basic panelized architecture proves out.

Reference Links

Sources used for manufacturing language, material stacks, welding approaches, and related foldable cooler construction.

- [Kelty Folding Cooler instructions](#): folding foam panels, Velcro assembly, removable liner.
- [SoftCoolers.com](#): fusion-welded liner and sweat-liner construction language.
- [Extremus Rendu Soft Cooler](#): TPU/nylon/EPE foam construction and induction heat welding claims.
- [Lovrix OEM cooler bag manufacturing](#): common fabric, foam, liner, webbing, welding, and leak-test options.
- [Polarthin flexible aerogel fabric](#): thin flexible insulation option for advanced hinge-zone testing.
- [Nutshell Cooler](#): foldable cooler using coconut-husk insulation as an alternate material story.